

NEXUS-1

Nuclear Plant Operator Console & Digital-Twin Platform

Operating Modes, Explained

What each of the six modes is for — in plain language

Written for a technical reader who is not a nuclear specialist

Sim · Training · Shadow · Advisory · Supervisory · Hybrid Digital Twin

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Build signature NX1-094C79E0A366D34F · June 2026

A SIMPLE GUIDE

NEXUS-1 is built around six operating modes. They are not six different apps — they are six ways the same console can be used, from a fully virtual plant you can experiment with freely, all the way to a live model that mirrors a real plant and forecasts where it is heading.

This guide explains, in everyday terms, what each mode is for and what you can and cannot do in it. The one rule that runs through all of them: the platform may watch and advise, but never directly controls a real reactor.

The six modes at a glance

Two are active today; four are on the roadmap. They build on one another.

Think of the modes as a ladder. At the bottom, everything is simulated and you can do anything safely. As you climb, the console connects to more of the real world — but with each step it does less touching and more watching, because acting on a real reactor is something only people and certified safety systems are allowed to do.

Mode	Status	In one line	Sees real data?	Controls real plant?
Simulation	Active	The whole plant, running as software	No — all simulated	No real plant
Training	Active	The simulator, with lessons and a score	No — all simulated	No real plant
Shadow	Roadmap	Watches a real plant, read-only	Yes	No
Advisory	Roadmap	Watches, and offers advice	Yes	No
Supervisory	Roadmap	Connected and coordinated, supervised	Yes	No
Hybrid Twin	Roadmap	A live model kept in step with reality	Yes	No (sim branch only)

The one rule behind every mode. NEXUS-1 can look at a real plant and it can give advice about it, but it never pushes the buttons. A real reactor’s safety controls stay separate, on dedicated hardware, operated by people. The software even enforces this for you: the moment the console is pointed at a real plant, the rod and SCRAM controls simply disappear from the screen.

Simulation

ACTIVE

The whole plant, running as software.

Everything you see is computed on the spot — the reactor, the gauges, the alarms. Nothing is wired to a real plant, so you can push it as hard as you like with zero consequences. This is the mode NEXUS-1 runs in today.

THINK OF IT LIKE... *a flight simulator, or a local dev environment loaded with realistic seed data — a safe place to pull every lever and watch what happens.*

YOU CAN

- ✓ Drive the reactor — move rods, trigger a SCRAM
- ✓ Run any scenario you can imagine
- ✓ Break things on purpose to see how it responds

YOU CAN'T

- ✗ Affect anything real — there is nothing real here

Training

ACTIVE

The simulator, with lessons and a score.

The same virtual plant, wrapped in guided exercises. You pick a drill, take the controls, and the system scores how well you hold the objective. Finished drills are written to the record for review.

THINK OF IT LIKE... *a driving-test course with an examiner, or a game's guided "tutorial mode" — structured practice with feedback.*

YOU CAN

- ✓ Run scored drills against a clear objective
- ✓ Control the live model and learn from each run
- ✓ Replay and review how you did

YOU CAN'T

- ✗ Touch a real plant — it is still entirely virtual

Shadow

ROADMAP

Watches a real plant, but can't touch it.

The console mirrors a live plant's real readings as they happen, but every control is taken away. Its whole job is to watch — and to flag the moments where the model's expectation drifts away from what the plant is actually doing. That drift is an early warning sign.

THINK OF IT LIKE... a “shadow” software deployment, or a read-only database replica — new code runs alongside production comparing results, but never serves live traffic. Or a trainee shadowing a senior: watching closely, hands off.

YOU CAN

- ✓ See live plant data in real time
- ✓ Spot where the model and reality disagree, early

YOU CAN'T

- ✗ Send any command
- ✗ Even reach the rod or SCRAM controls — they are hidden

Advisory

ROADMAP

Watches, and offers advice — people decide.

Everything Shadow does, plus a layer of recommendations: predicted maintenance issues, an optimisation suggestion, a plain-language assistant that explains what it is seeing. All of it is advice. A human makes every decision.

THINK OF IT LIKE... a GPS suggesting a route, or a code-review bot that flags issues and proposes a fix — genuinely useful, but it never merges the change itself.

YOU CAN

- ✓ See predictions and recommendations laid out clearly
- ✓ Accept the advice, or ignore it

YOU CAN'T

- ✗ Let the system act on its own — it never actuates

Supervisory

ROADMAP

Connected across the system, under human supervision.

Wired into the wider control infrastructure for monitoring and coordination, with full security hardening around it. It oversees and coordinates the bigger picture — but a person still signs off on anything that matters, and it still has no direct hold on the reactor.

THINK OF IT LIKE... *a mission-control room or a network operations centre: every screen is connected and coordinated, yet a human authorises the actions.*

YOU CAN

- ✓ Monitor and coordinate across the connected system
- ✓ Operate with full security controls in place

YOU CAN'T

- ✗ Skip the human in the loop
- ✗ Take direct control of the reactor

Hybrid Digital Twin

ROADMAP

A live model that keeps in step with the real plant.

The simulation runs continuously next to the real plant and is constantly nudged by live data so the two stay in step. From that live state you can branch “what-if” forecasts — where is this heading if nothing changes, or if we do X?

THINK OF IT LIKE... *live-traffic navigation that re-routes as conditions change, or a weather model fed by live sensors — a model that mirrors reality and predicts from it.*

YOU CAN

- ✓ Watch the model track the live plant
- ✓ Run what-if forecasts from the current real state
- ✓ Command the simulation branch freely

YOU CAN'T

- ✗ Command the real plant — the live side stays hands-off

How it all fits together

One console, six gears

The simplest way to hold all six in your head: it is one console with six gears. The lowest gears (Simulation and Training) are fully virtual and completely safe to experiment with — that is where NEXUS-1 lives today. The higher gears (Shadow, Advisory, Supervisory, Hybrid) are the roadmap: each one connects the console to more of the real world, and each one is built on top of the gear below it.

Notice the pattern as you go up the gears: the platform gets more connected but does less acting. Shadow only watches. Advisory adds suggestions. Supervisory coordinates the wider picture. Hybrid keeps a live model running in step. At no point does the software take the wheel of a real reactor — that always stays with people and with dedicated safety hardware. It is a deliberate design choice, and the interface enforces it.

A note on honesty

NEXUS-1 is a portfolio demonstration, not a product connected to any real plant. The four roadmap modes are shown as clearly-labelled concept demonstrations: they illustrate how the console would behave, using the built-in simulation in place of a live feed. Saying that plainly is the point — it is what keeps the demonstration trustworthy.

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